

$$5^{2^{l-3}} \equiv 1 + 2^{l-1} \pmod{2^l}$$

$$\phi(2^l) = 2^l - 2^{l-1} = 2^{l-1}$$

$$5^{2^{l-1}} \equiv 1 \pmod{2^l}$$

$$5^{2^{l-1}} - 1 \equiv 0 \pmod{2^l}$$

$$(5^{2^{l-2}} - 1)(5^{2^{l-2}} + 1) \equiv 0 \pmod{2^l}$$

$$(5^{2^{l-3}} - 1)(5^{2^{l-3}} + 1)(5^{2^{l-2}} + 1) \equiv 0 \pmod{2^l}$$

$$V_2(5^{2^{l-3}} - 1) + V_2(5^{2^{l-3}} + 1) + V_2(5^{2^{l-2}} + 1)$$

$$\geq 1$$

$$5^{2^{l-3}} + 1 \equiv 1^{2^{l-3}} + 1 \equiv 2 \pmod{4}$$

$$5^{2^{l-2}} + 1 \equiv 1^{2^{l-2}} + 1 \equiv 2 \pmod{4}$$

$$2 \nmid 5^{2^{l-3}} - 1 \quad \wedge \quad 2 \nmid 5^{2^{l-2}} - 1 \quad \wedge \quad 2 \nmid 1$$

$$\Rightarrow V_2(5^{2^{l-2}} - 1) = V_2(5^{2^{l-3}} - 1) = 1$$

$$\Rightarrow V_2(5^{2^{l-3}} - 1) \geq l-2$$

$$5^{2^{l-3}} - 1 \equiv 0 \pmod{2^{l-2}}$$

$$2^{l-2} \mid 5^{2^{l-3}} - 1$$

$$5^{2^{l-3}} - 1 = 2^{l-2} \cdot d \quad d \in \mathbb{Z}$$

$$2^{l-2} \cdot d = 5^{2^{l-3}} - 1 \equiv 2^{l-1} \pmod{2^l}$$

$$2^{l-2} d \equiv 2^{l-1} \pmod{2^l}$$

$$\Leftrightarrow d \equiv 2 \pmod{4}$$

$$V_2(5^{2^{l-3}} - 1) = l-1$$

LZW

$$V_2(5 \cdot 1) = 0$$

$$V_2(5-1) = 2 \geq 2$$

$$\begin{aligned} V_2(5^{2^{l-3}} - 1) &= V_2(5-1) + V_2(2^{l-3}) \\ &= 2 + l-3 = l-1 \end{aligned}$$

$$\stackrel{\equiv}{\equiv} \begin{array}{l} 2 \nmid b \wedge \\ b \in \mathbb{Z} \end{array} \quad 5^{2^{l-3}} - 1 = 2^{l-1} \cdot b$$

$$2^{l-1} \cdot b \equiv 2^{l-1} \pmod{2^l}$$

$$2 \nmid b \Rightarrow b = 2k+1 \quad \text{gdzie } k \in \mathbb{Z}$$

$$\begin{aligned} 2^{l-1} \cdot b &\equiv 2^{l-1} \cdot (2k+1) \equiv 2^L \cdot k + 2^{l-1} \\ &\equiv 2^{l-1} \pmod{2^L} \end{aligned}$$

$$\text{ord}_{2^l}(5) = 2^{l-2}$$

$$5^{2^{l-2}} \equiv 1 \pmod{2^l}$$

$$5^{2^{l-3}} \equiv 1 + 2^{l-4} \pmod{2^l}$$

$$\Leftrightarrow 5^{2^{l-2}} \equiv 1 + 2^l + 2^{2l-2} \equiv 1$$

$$\text{ord}_{2^l}(5) \leq 2^{l-2}$$

$$\text{ord}_{2^l}(5) \mid \phi(2^l) = 2^{l-1}$$

$$\Rightarrow \text{ord}_{2^l}(5) = 2^k \text{ gdzie}$$

$$0 \leq k \leq l-2$$

$$k \neq l-2$$

$$5^{2^k} \equiv 1 \pmod{2^l}$$

$\Downarrow$

$$l \leq V_2(5^{2^k} - 1) = V_2(4) + V_2(2^k) = k + 2$$

$$l \leq k+2 \Leftrightarrow k \geq l-2 \quad \Downarrow$$

$$\Rightarrow k = l-2$$

$$\Rightarrow \text{ord}_2(5) = 2^{l-2}$$

